

$$\begin{cases} \frac{\partial u}{\partial t} - \nu \frac{\partial^2 u}{\partial x^2} = f & \bar{a} < x < \bar{b} \end{cases}$$

$$\begin{cases} \nu \frac{\partial u}{\partial x}(\bar{a}) = \nu \frac{\partial u}{\partial x}(\bar{b}) = 0 & (N) \end{cases}$$

$$\begin{cases} u(x, 0) = u_0(x) & \bar{a} < x < \bar{b} & (I) \end{cases}$$

$$\int_{\bar{a}}^{\bar{b}} \frac{\partial u}{\partial t} \nu dx + \underbrace{\int_{\bar{a}}^{\bar{b}} \nu \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} dx}_{a(u, v)} - \left[\nu \frac{\partial u}{\partial x} v \right]_{\bar{a}}^{\bar{b}} = \int_{\bar{a}}^{\bar{b}} f v dx$$

$$a(v, v) = \int_{\bar{a}}^{\bar{b}} \nu \left(\frac{\partial v}{\partial x} \right)^2 dx = \nu \left\| \frac{\partial v}{\partial x} \right\|_{L^2(\bar{a}, \bar{b})}^2$$

$$\nu = \text{const} > 0$$

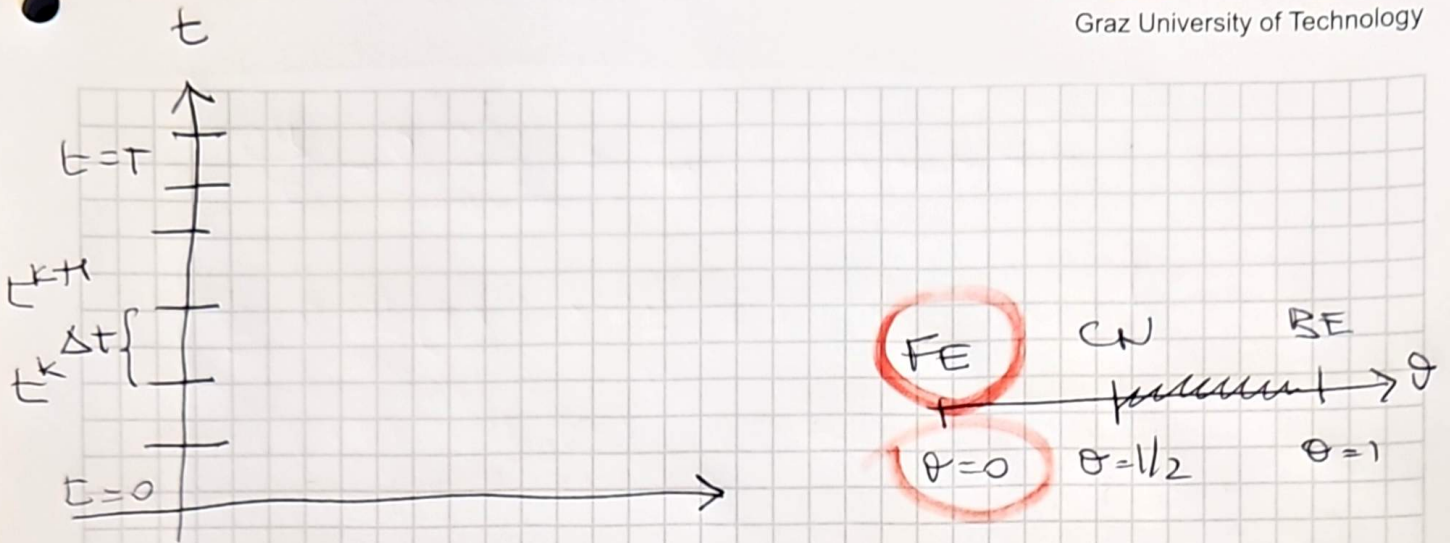
$$V = H^1(\bar{a}, \bar{b}) \quad \left(\Gamma_D = \{\emptyset\} \right)$$

$$\|v\| = \left(\|v\|_{L^2(\bar{a}, \bar{b})}^2 + \left\| \frac{\partial v}{\partial x} \right\|_{L^2(\bar{a}, \bar{b})}^2 \right)^{1/2}$$

NOT COERCIVE!! $a(v, v) \geq \alpha \|v\|^2$ NOT TRUE

$$\begin{aligned} a(v, v) + \lambda \|v\|_{L^2(\bar{a}, \bar{b})}^2 &= \nu \left\| \frac{\partial v}{\partial x} \right\|_{L^2(\bar{a}, \bar{b})}^2 + \lambda \|v\|_{L^2(\bar{a}, \bar{b})}^2 \\ &\geq \min(\nu, \lambda) \left(\left\| \frac{\partial v}{\partial x} \right\|_{L^2(\bar{a}, \bar{b})}^2 + \|v\|_{L^2(\bar{a}, \bar{b})}^2 \right) \geq \min(\lambda, \nu) \|v\|^2 \end{aligned}$$

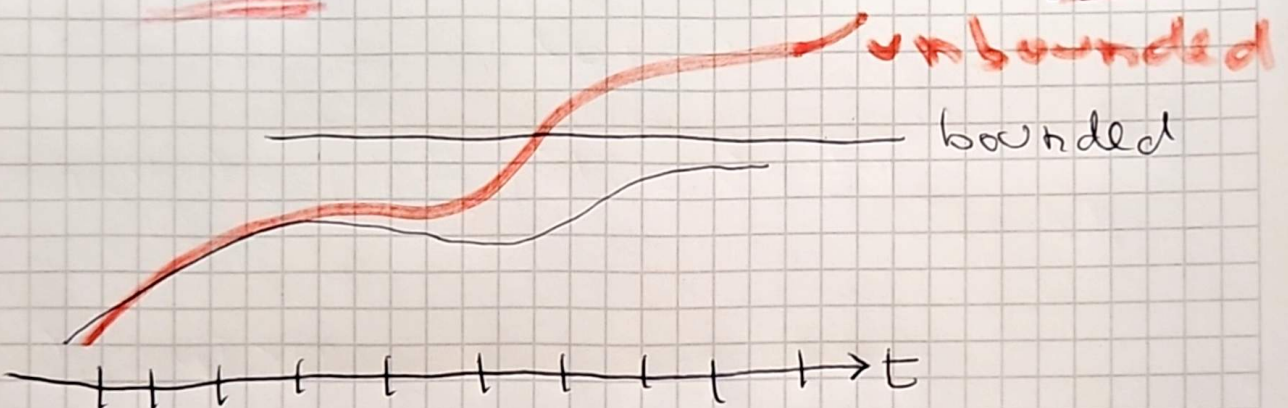
coercive



FE, BE:
CN ($\theta=1/2$)

error:
"

$O(\Delta t)$ 1st order
 $O(\Delta t^2)$ 2nd order



Condition on Δt to guarantee stability,

Conditional stability

$$0 \leq \theta < 1/2$$

$$0 < \Delta t \leq c h^2$$

$$\theta \geq 1/2$$

$0 < \Delta t$ arbitrary!
Unconditionally stability

$$\underbrace{\int_{\Omega} u_h^{kH} u_h^{kH}}_{\text{CS}} + \Delta t \underbrace{a(u_h^{kH}, u_h^{kH})}_{\propto \|u_h^{kH}\|_{\sqrt{\cdot}}^2} = \int_{\Omega} u_h^k u_h^{kH}$$

$$\|u_h^{kH}\|_{L^2(\Omega)}^2 + \Delta t \alpha \|u_h^{kH}\|_{\sqrt{\cdot}}^2 \leq \underbrace{\|u_h^k\|_{L^2(\Omega)}^2}_A \underbrace{\|u_h^{kH}\|_{L^2(\Omega)}^2}_B$$

$$AB \leq \frac{1}{2} A^2 + \frac{1}{2} B^2 \quad \text{Young inequality } (\varepsilon = 1/2)$$

$$\Uparrow$$

$$AB \leq \varepsilon A^2 + \frac{1}{4\varepsilon} B^2 \quad \forall \varepsilon > 0$$

$$\leq \frac{1}{2} \|u_h^k\|_{L^2(\Omega)}^2 + \frac{1}{2} \|u_h^{kH}\|_{L^2(\Omega)}^2$$

$$\downarrow$$

$$\cancel{\frac{1}{2}} \|u_h^{kH}\|_{L^2(\Omega)}^2 + 2\Delta t \alpha \|u_h^{kH}\|_{\sqrt{\cdot}}^2 \leq \cancel{\frac{1}{2}} \|u_h^k\|_{L^2(\Omega)}^2$$

new time-step t^{kH} previous t^k

$$\Rightarrow \underbrace{\left(\|u_h^{kH}\|_{L^2}^2 - \|u_h^k\|_{L^2}^2 \right)}_{\text{Telescopic sum}} + 2\Delta t \alpha \|u_h^{kH}\|_{\sqrt{\cdot}}^2 \leq 0$$

$$\sum_{k=0}^n \left(\|u_h^{kH}\|_{L^2}^2 - \|u_h^k\|_{L^2}^2 \right) + \sum_{k=0}^n 2\Delta t \alpha \|u_h^{kH}\|_{\sqrt{\cdot}}^2 \leq 0 \quad \forall k = 0, \dots, M-1$$

telescopic sum: $\sum_{k=0}^{n-1} (c_{k+1} - c_k) =$

$$= (\cancel{c_1} - \cancel{c_0}) + (\cancel{c_2} - \cancel{c_1}) + (\cancel{c_3} - \cancel{c_2}) + \dots + (\cancel{c_n} - \cancel{c_{n-1}})$$

$$= c_n - c_0$$

$$\left(\|u_h^{n+1}\|_{L^2(\Omega)}^2 - \|u_h^0\|_{L^2(\Omega)}^2 \right)$$

$$+ 2\alpha \Delta t \sum_{k=0}^n \|u_h^{k+1}\|_{L^2(\Omega)}^2 \leq 0$$

BE

$$\Rightarrow \|u_h^{n+1}\|_{L^2(\Omega)}^2 + 2\alpha \Delta t \sum_{k=0}^n \|u_h^{k+1}\|_{L^2(\Omega)}^2 \leq \|u_h^0\|_{L^2(\Omega)}^2$$

a Norm for u_h^{n+1}

norm of the initial data

$\Rightarrow$ boundedness

$\Rightarrow$ stability

no assumption on Δt

$\Rightarrow$ unconditional stability of BE!

About Young inequality:

$$0 \leq \underbrace{(\sqrt{\varepsilon} a)}_X - \underbrace{\frac{1}{2\sqrt{\varepsilon}} b}_Y \Big)^2 = X^2 + Y^2 - 2XY$$

$$= \varepsilon a^2 + \frac{1}{4\varepsilon} b^2 - \cancel{2\sqrt{\varepsilon} a} \cancel{\frac{1}{2\sqrt{\varepsilon}} b}$$

$$\Rightarrow \boxed{ab \leq \varepsilon a^2 + \frac{1}{4\varepsilon} b^2} \quad \forall \varepsilon > 0$$

Young inequality.

$\forall k$

$$\begin{aligned} \|u_h^k\|_{L^2} &\leq \frac{1}{\sqrt{1+2\alpha\Delta t}} \|u_h^k\|_{L^2} \\ &\leq \frac{1}{\sqrt{1+2\alpha\Delta t}} \|u_h^{k-1}\|_{L^2} \\ &\leq \dots \leq \left(\frac{1}{\sqrt{1+2\alpha\Delta t}} \right)^{kM} \|u_h^0\|_{L^2} \end{aligned}$$

$\Rightarrow$

~~for~~ $\forall 1 \leq k \leq M$: stability

$$\boxed{\|u_h^k\|_{L^2} \leq \left[\frac{1}{\sqrt{1+2\alpha\Delta t}} \right]^k \|u_h^0\|_{L^2}}$$

$\rightarrow 0$ if $k \rightarrow +\infty$

Question about interpretation - f
weak coercivity

$$\frac{\partial u}{\partial t} - \nu \frac{\partial^2 u}{\partial x^2} = f$$

Neumann conditions

$$a(u, v) = \int_{\Omega} \nu \frac{\partial u}{\partial x} \frac{\partial v}{\partial x}$$

weakly coercive
(not strongly coercive).

$$u = e^{xt} w \quad \text{new variable} \quad w = e^{-xt} u$$

$$\frac{\partial}{\partial t} (e^{xt} w) - \nu \frac{\partial^2}{\partial x^2} (e^{xt} w) = f$$

$$\lambda e^{xt} w + e^{xt} \frac{\partial w}{\partial t} - e^{xt} \nu \frac{\partial^2 w}{\partial x^2} = f$$

$$\Rightarrow \frac{\partial w}{\partial t} - \nu \frac{\partial^2 w}{\partial x^2} + \lambda w = \underbrace{e^{-xt} f}_{g(t) \text{ known}}$$

$$\tilde{a}(w, v) = \underbrace{\int_{\Omega} \nu \frac{\partial w}{\partial x} \frac{\partial v}{\partial x}}_{a(w, v)} + \underbrace{\lambda \int_{\Omega} w v}_{\lambda \|w\|_{L^2}^2}$$

$$\tilde{a}(v, v) = a(v, v) + \lambda \|v\|_{L^2}^2 \geq \alpha \|v\|_{L^2}^2$$

↑
weakly coercive

$\tilde{a}$ is coercive!

(21) σ -method

Eigenvalues of a matrix B :

$$\boxed{B \vec{w} = \lambda \vec{w}} \quad (\lambda, \vec{w}) \text{ eigenvalue / eigenvector of } B$$

Generalized eigenvalues:

$$\boxed{B \vec{w} = \lambda M \vec{w}} \quad (\lambda, \vec{w}) \text{ generalized eigenvalue / eigenvector}$$

M pos. def. matrix & symm. of B, M .

Assume B is SPD (symmetric & positive definite)

$\Rightarrow$ all its eigenvalues / generalized eigenvalues are strictly positive

B, M ($N \times N$) $\rightarrow$ N eigenvalues.

GENERALIZE THIS TO symm. hermite bilinear forms!

$a(\cdot, \cdot)$ symm hermite bil. form in $V \times V$

we call (λ, w) eigenvalue and eigenfunction of $a(\cdot, \cdot)$ if

$$\lambda \in \mathbb{R}, w \in V: \quad a(w, v) = \lambda (w, v) \quad \forall v \in V$$

$$= \lambda \int_{\Omega} w v$$

INFINITELY MANY! eigenvalues unbounded ($\nearrow \infty$)

Replace V by V_h :

(X) ? $\lambda_h \in \mathbb{R}, > 0, w_h \in V_h: a(w_h, v_h) = \lambda_h (w_h, v_h) \quad \forall v_h \in V_h$
(2.3) $\Rightarrow (\lambda_h, w_h)$ provide an approximation of (λ, w)

$$w_h(x) = \sum w_j \varphi_j(x) \quad \{\varphi_j\} \text{ basis for } V_h$$

$$v_h = \varphi_i$$

$$\Rightarrow a\left(\sum_j w_j \varphi_j, \varphi_i\right) = \lambda_h \left(\sum_j w_j \varphi_j, \varphi_i\right) \quad \forall i$$

$$\Rightarrow \sum_j w_j a(\varphi_j, \varphi_i) = \sum_j w_j \lambda_h (\varphi_j, \varphi_i)$$

$$\Rightarrow \begin{matrix} A_{ij} \\ \boxed{A \vec{w} = \lambda_h M \vec{w}} \\ \text{Generalized} \\ \text{eigenvalue problem} \\ N_h \text{ eigenvalues / eigenfunctions} \end{matrix} \Rightarrow \vec{w} = \begin{matrix} M_{ij} \\ \begin{bmatrix} w_1 \\ \vdots \\ w_{N_h} \end{bmatrix} \end{matrix}$$

Let $\{w_h^j, \lambda_h^j\}$ be the N_h eigenvalues / eigenfunctions of (X)

We can assume that $\{w_h^j\}$ are normalized, that is

$$(XX) \int_{\Omega} w_h^j w_h^i = \delta_{ij} = \begin{cases} 1 & \text{if } j = i \\ 0 & \text{if } j \neq i \end{cases}$$

$$\Rightarrow \|w_h^j\|_2 = \int_{\Omega} (w_h^j)^2 = 1 \quad \text{unitary norm in } L^2$$

let now $\forall v_h \in V_h, \quad x_h(x) = \sum_{j=1}^{N_h} v_j \underbrace{w_h^j(x)}_{\text{basis of } V_h}$

N_h orthonormal $\Rightarrow$
 N_h linearly independent
 $\Rightarrow \{w_h^j(x)\}_{j=1}^{N_h}$ basis of V_h

Go back to (*) and take w_h^i :
 $\Rightarrow a(\sum$

Go back to the θ -method: (21) $F=0$

$$\int_{\Omega} \frac{u_h^{k+1} - u_h^k}{\Delta t} v_h + a(\theta u_h^{k+1} + (1-\theta) u_h^k, v_h) = 0$$

$$u_h^k(x) = \sum_{j=1}^{N_h} u_j^k \underbrace{w_h^j(x)}_{\text{basis}}, \quad \text{set } v_h = w_h^i$$

↑
unknown coefficients at time t^{k+1}

$$\begin{aligned} \rightarrow \int_{\Omega} \frac{1}{\Delta t} \left(\sum_j (u_j^{k+1} - u_j^k) w_h^j \right) w_h^i &+ a \left(\theta \sum_j u_j^{k+1} w_h^j + (1-\theta) \sum_j u_j^k w_h^j, w_h^i \right) = 0 \quad \forall i \end{aligned}$$

$$\frac{1}{\Delta t} \sum_j (u_j^{kH} - u_j^k) \underbrace{\int \omega_h^j \omega_h^i}_{S_{ij}} + \sum_j (\theta u_j^{kH} + (1-\theta) u_j^k) \cdot \underbrace{\alpha(\omega_h^j, \omega_h^i)}_{=0}$$

Thanks to (23):
 $x_h^j(\omega_h^j, \omega_h^i)$

$$\Rightarrow \frac{1}{\Delta t} (u_i^{kH} - u_i^k) + \sum_j (\theta u_j^{kH} + (1-\theta) u_j^k) \underbrace{x_h^j(\omega_h^j, \omega_h^i)}_{S_{ij}} = 0$$

$$\Rightarrow \frac{1}{\Delta t} (u_i^{kH} - u_i^k) + [\theta u_i^{kH} + (1-\theta) u_i^k] x_h^i = 0, i=1, \dots, N_h$$

$$\Rightarrow u_i^{kH} + \Delta t \theta u_i^{kH} x_h^i = u_i^k - \Delta t (1-\theta) x_h^i u_i^k$$

$$\Rightarrow [1 + \Delta t \theta x_h^i] u_i^{kH} = [1 - \Delta t (1-\theta) x_h^i] u_i^k$$

$$\Rightarrow u_i^{kH} = \underbrace{\left(\frac{1 - \Delta t (1-\theta) x_h^i}{1 + \Delta t \theta x_h^i} \right)}_{\rho_i} u_i^k \quad \forall i=1, \dots, N_h$$

Assume $|\rho_i| < 1 \quad \forall i=1, \dots, N_h$

$$\Rightarrow |u_i^{kH}| < |u_i^k| \quad \forall i, \forall k$$

$$\Rightarrow \forall n=1, \dots, N_h \quad |u_i^n| < |u_i^0| \quad \forall i=1, \dots, N_h$$

$$\Rightarrow \sum_{i=1}^{N_h} |u_i^n|^2 < \sum_{i=1}^{N_h} |u_i^0|^2$$

$t_0 \quad t_1^* \quad t^2 \quad \dots \quad t^n \quad t$

Remember that:

$$u_h^n(x) = \sum_{j=1}^{N_h} u_j^n w_h^j(x)$$

↓

$$\begin{aligned} \|u_h^n\|_{L^2(\Omega)}^2 &= \int_{\Omega} (u_h^n)^2 = \int_{\Omega} \left(\sum_j u_j^n w_h^j \right) \left(\sum_i u_i^n w_h^i \right) \\ &= \sum_j \sum_i u_j^n u_i^n \underbrace{\int_{\Omega} w_h^j w_h^i}_{\delta_{ij}} \\ &= \sum_j u_j^n u_j^n = \sum_j (u_j^n)^2 \end{aligned}$$

$$\Rightarrow \sum_{j=1}^{N_h} (u_j^n)^2 = \|u_h^n\|_{L^2(\Omega)}^2$$

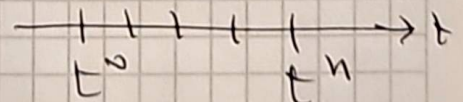
From (..) we deduce that:

$$\|u_h^n\|_{L^2(\Omega)}^2 < \|u_{oh}\|_{L^2(\Omega)}^2 \quad \forall n$$

⇒

$$\boxed{\|u_h^n\|_{L^2(\Omega)} < \|u_{oh}\|_{L^2(\Omega)} \quad \forall n=1, \dots, M_h}$$

stability!



under the assumption that $|g_i| < 1 \quad \forall i$

$$|p_i| = \left| \frac{1 - \Delta t(1-\theta)\lambda_h^i}{1 + \Delta t\theta\lambda_h^i} \right| < 1 \quad \forall i$$

$$\Leftrightarrow -1 < p_i < 1 \quad \forall i$$

$$\Leftrightarrow -1 - \theta\lambda_h^i \Delta t < 1 - (1-\theta)\lambda_h^i \Delta t < 1 + \theta\lambda_h^i \Delta t$$

$$\theta > 0, \lambda_h^i > 0, \Delta t > 0$$

$$\Leftrightarrow -\frac{2}{\lambda_h^i \Delta t} - \theta < \theta - 1 < \theta$$

↑
always true

$$\Downarrow$$

$$2\theta - 1 > -\frac{2}{\lambda_h^i \Delta t}$$

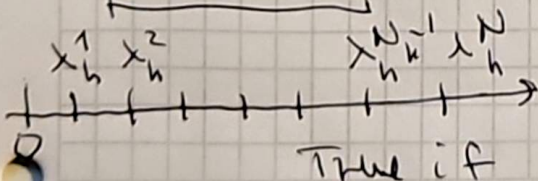
$\underbrace{\hspace{10em}}_{< 0} \Rightarrow \text{always TRUE}$
 $\underbrace{\hspace{10em}}_{> 0}$
 $\forall \Delta t > 0$

if $\theta \geq 1/2$

$\Rightarrow$ unconditional stability (CN, BE)

if $\theta < 1/2$

$$\Rightarrow \Delta t < \frac{2}{(1-2\theta)\lambda_h^i} \quad \forall i$$



$$\Delta t < \frac{2}{(1-2\theta)\lambda_h^N} \approx \frac{2}{(1-2\theta)} h^2$$